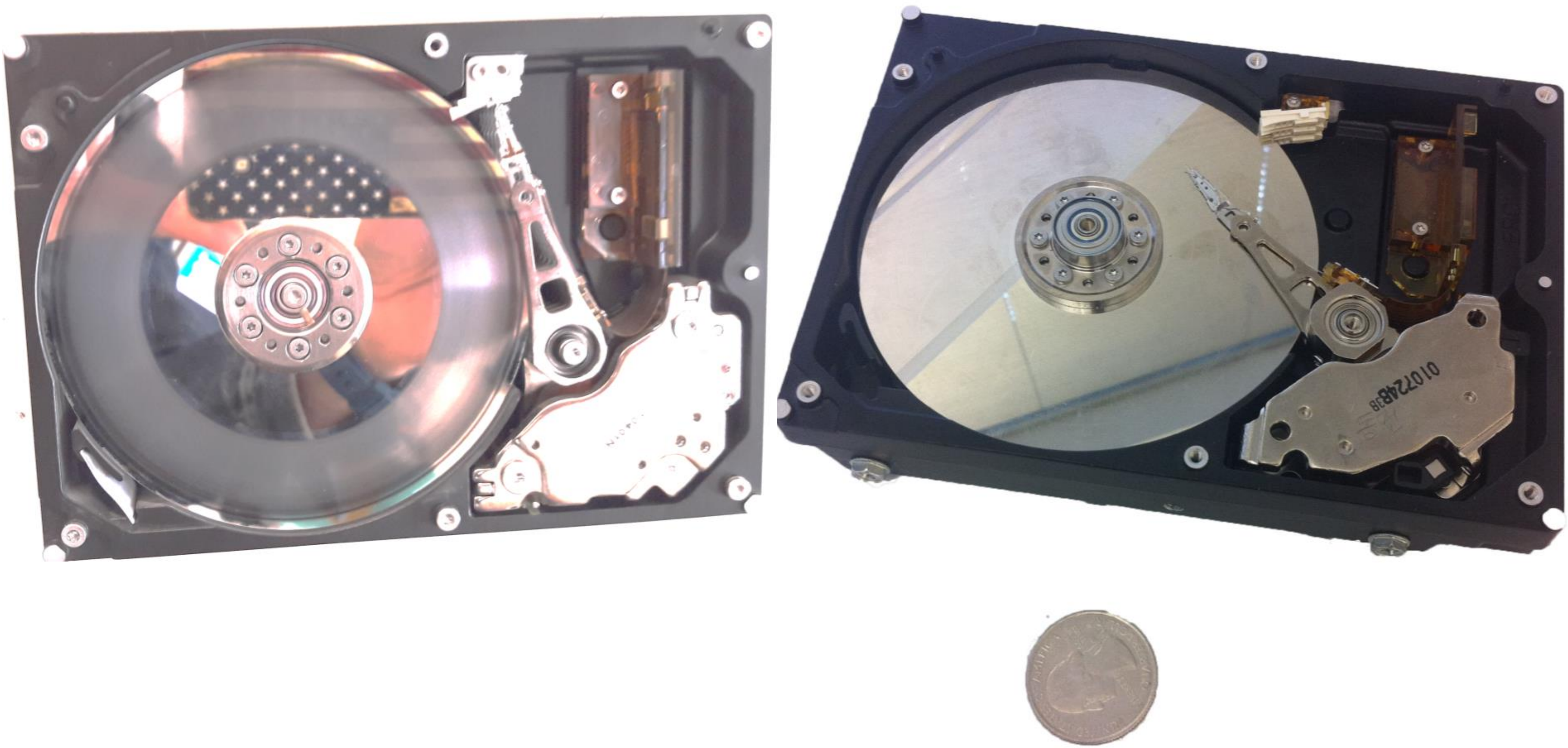
Disks

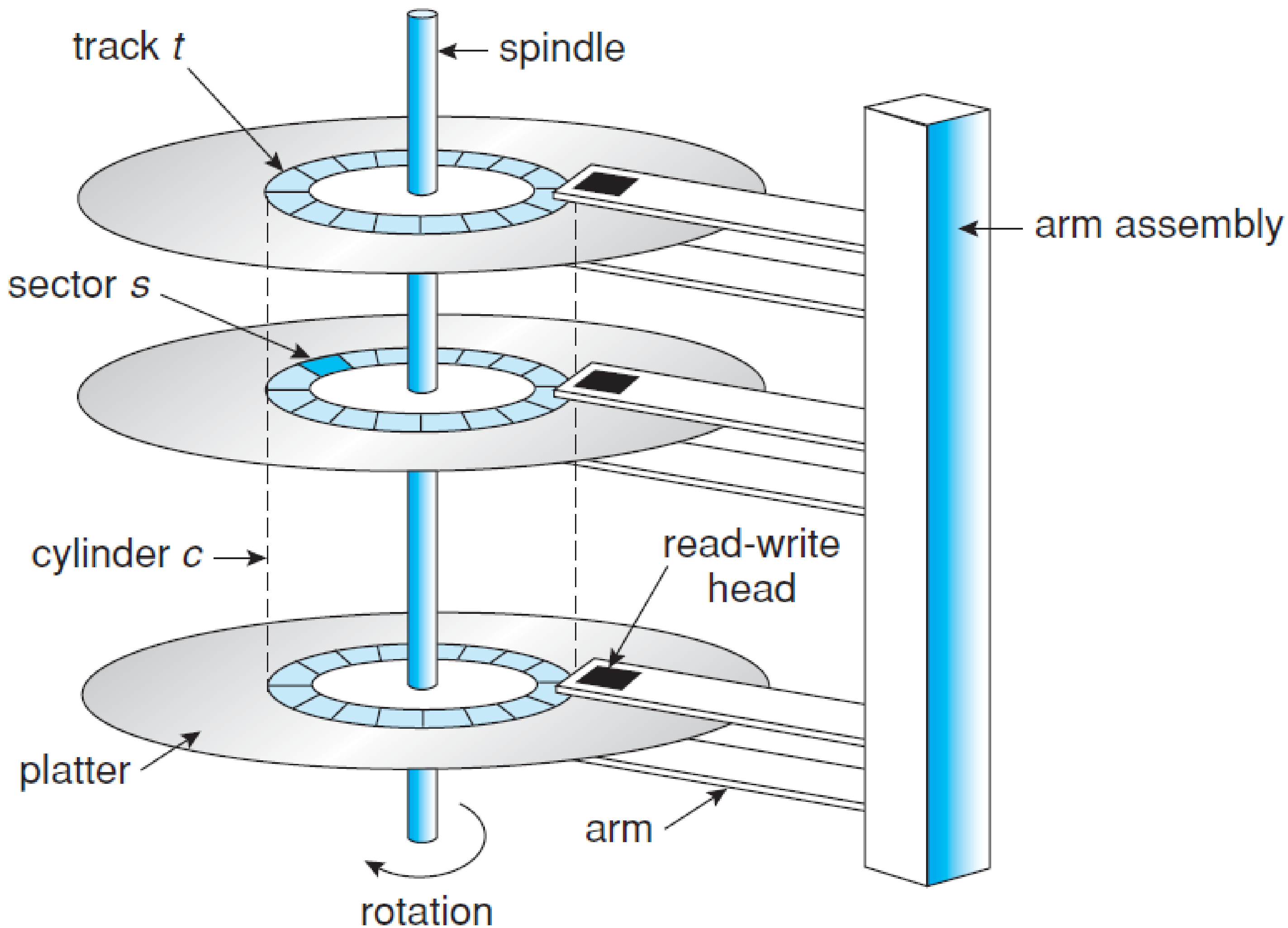
Storage Devices

- Magnetic disks
  - Storage that rarely becomes corrupted
  - Large capacity at low cost
  - Block level random access
  - Slow performance for random access
  - Better performance for streaming access
- Flash memory
  - Storage that rarely becomes corrupted
  - Good capacity at intermediate cost (2 x disk)
  - Block level random access
  - Good performance for reads; worse for random writes

Magnetic Disk



Magnetic Disk Components



Disk Tracks

- ~ 1 micron wide
  - Wavelength of light is ~ 0.5 micron
  - Resolution of human eye: 50 microns
  - 100K on a typical 2.5” disk
- Separated by unused guard regions
  - Reduces likelihood neighboring tracks are corrupted during writes (still a small non-zero chance)
- Track length varies across disk
  - Outside: More sectors per track, higher bandwidth
  - Disk is organized into regions of tracks with same # of sectors/track
  - Only outer half of radius is used
    - Most of the disk area in the outer regions of the disk

Sectors

Sectors contain sophisticated error correcting codes

- Disk head magnet has a field wider than track
- Hide corruptions due to neighboring track writes

- Sector sparing
  - Remap bad sectors transparently to spare sectors on the same surface
- Slip sparing
  - Remap all sectors (when there is a bad sector) to preserve sequential behavior
- Track skewing
  - Sector numbers offset from one track to the next, to allow for disk head movement for sequential ops

Sectors & blocks

- **At low level** the controller accesses individual sectors
  - Typical sector size is 256/512 bytes
  - Identified by a triple: <#cylinder, #face, #sector>
- **At the higher level**, the disk driver groups a *set of contiguous sectors in a block*
  – Typical block size is 2/4/8 Kbytes
  – identified by a pointer on a contiguous address space (from 0 to max-blocks)

Sectors & blocks

Given a sector number b, and a triple $< c, f, s >$:

$$b = c(\#faces \cdot \#sectors) + f(\#sectors) + s$$

* assumption 1 block corresponds to 1 sector

- $\#faces$ is the number of faces in the disk
- $\#sectors$ is the number of sectors per track

Consequently:

$$c = b\ div\ (\#faces \cdot \#sectors)$$

$$f = \big(b\ mod\ (\#faces \cdot \#sectors)\big) div\ \#sectors$$

$$s = \big(b\ mod\ (\#faces \cdot \#sectors)\big)\ mod\ \#sectors$$

Example

Disk with 100 cylinders, 4 faces, 2000 sectors per track.

A file uses the blocks 94421,94422,94423. Which is the <c,f,s> for each block?

c= 94421 div (4*2000) = 11 (remainder 6421)

f= 6421 div 2000 = 3 (remainder 421)

s= 421

Therefore, the triples are: <11,3,421> <11,3,422> <11,3,423>

Disk Performance

Disk Latency= Seek Time + Rotation Time + Transfer Time

- **Seek Time**: time to move disk arm over track (1-20ms)
  - Fine-grained position adjustment necessary for head to “settle”
  - Head switch time ~ track switch time (on modern disks)
- **Rotation Time**: time to wait for disk to rotate under disk head
  - Disk rotation: 4 – 15ms (depending on price of disk)
- **Transfer Time**: time to transfer data onto/off of disk
  - Disk head transfer rate: 50-100MB/s (5-10 usec/sector)
  - Host transfer rate dependent on I/O connector (USB, SATA, …)

Disk Information (2008)

Size	
Platters/heads	2/4
Capacity	320 GB
Performance	
Spindle speed	7200 RPM
Average seek time read/write	10.5 ms / 12.0 ms
Maximum seek time	19 ms
Track-to-track seek time	1 ms
Transfer rate (surface to buffer)	54-128 MB/s
Transfer rate (buffer to host)	375 MB/s
Buffer memory	16 MB
Power	
Typical	16.35 W
Idle	11,68 W

Question

- How long to complete 500 random disk reads, in FIFO order?

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- How long to complete 500 random disk reads, in FIFO order?
  - Seek: average 10.5 msec
  - Rotation: average 4.15 msec
  - Transfer: 5-10 usec
- 500 * (10.5 + 4.15 + 0.01)/1000 = 7.3 seconds

Question

- How long to complete 500 sequential disk reads?

Question

- How long to complete 500 sequential disk reads?
  - Seek Time: 10.5 ms (to reach first sector)
  - Rotation Time: 4.15 ms (to reach first sector)
  - Transfer Time: (outer track)

    500 sectors * 512 bytes / 128MB/sec = 2ms

Total: 10.5 + 4.15 + 2 = 16.7 ms

Might need an extra head or track switch (+1ms)

Track buffer may allow some sectors to be read off disk out of order (-2ms)

Question

Disk with 100 cylinders, 4 faces, 2000 sectors per track.

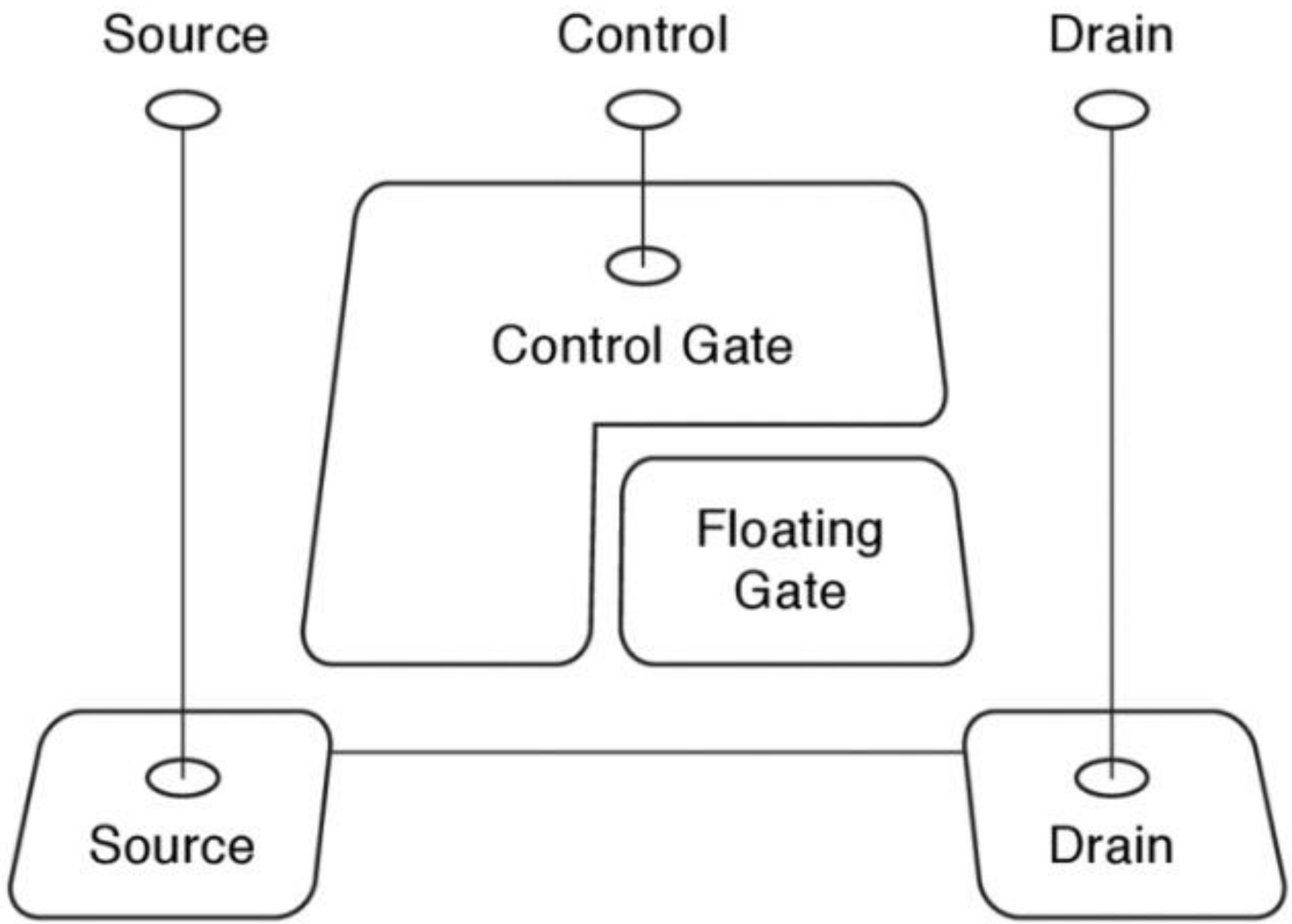
A file uses blocks 94421,94422,94423 (<11,3,421> <11,3,422> <11,3,423>). A sector is read in 0.01ms, the seek time between two consecutive tracks is 0.01ms, and the average time to reach the desired sector is half the rotation time. Considering that the arm is at cylinder 22, and that the DMA controller has enough buffers, compute the time to read the file blocks.

(22-11)*0.01 + 20/2 + 3*0.01 = 10.14ms

SSD Disk

- A Solid State Drive (SSD) is a non-volatile storage device based on flash-memory technology
- SSDs are more reliable and faster than hard disks (HDs) because they have no moving parts and no seek time
  - SSDs commonly use a simple FCFS scheduling policy.
- They consume less power and are more expensive per MB of data. The capacity of HD is usually larger.
- In some systems, SSDs are used as an additional cache tier in the memory hierarchy

Flash Memory in a nutshell



Flash Memory

- Writes must be to “clean” cells; no update in place
  - Large block erasure required before write
  - Erasure block: 128 – 512 KB
  - Erasure time: Several milliseconds
- Write/read page (2-4KB)
  - 50-100 usec

SSD disk information (2011)

Size	
Capacity	300 GB
Page size	4 KB
Performance	
Bandwidth (sequential reads)	270 MB/s
Bandwidth (sequential writes)	210 MB/s
Read/write latency	75 μs
Random reads per second	38,500
Random writes per second	2,000 (2,400 with 20% space reserve)
interface	SATA 3GB/s
Endurance	
Endurance	1.1 PB (1.5 PB with 20% space reserve)
Power	
Power consumption(active/idle)	3.7 W / 0.7 W

Question

- Why are random writes so slow?
  - Random write: 2000/sec
  - Random read: 38500/sec

Flash Translation Layer

- To avoid the cost of an erase for each write, pages are erased in advance
  - Clean pages always available for a new write
- This means that a write cannot be directed to an arbitrary page in the memory, but only to one previously erased page
- What happens if you rewrite a block of a file?
  - The page storing the block cannot be rewritten immediately…
  - It need to be erased first but with surrounding pages!
  - A clean page is used to apply the write, but this page is somewhere in the disk…
    - The old page goes to garbage for recycling
- How to know where the pages of my file are?

Flash Translation Layer

- Flash device firmware maps logical page # to a physical location
  - Move live pages as needed for erasure
    - Garbage collect empty erasure block by copying live pages to new location
  - Wear-levelling
    - Can only write each physical page a limited number of times
  - Avoid pages that no longer work
- Transparent to the device user

File System – Flash

- File systems on magnetic disks do not need to tell the disk what blocks are in use:
  - When a block is no longer used it is marked free in the bitmap
  - The file system reuse it whenever it wants
- When these FS were used first on flash drives the performances decayed dramatically over time

File System – Flash

- The Flash Translation Layer got busy on garbage collection:
  - Live blocks must be remapped to a new location …
  - … to compact the free pages in order to proceed with block erasure
- This even with a large amount of free space
  - For example, if the FS move a large file from a range of blocks to another one …
  - … the storage does not know that the old blocks can go to garbage …
  - … unless the FS tells it!

File System – Flash

- TRIM command
  - File system tells device when pages are no longer in use
  - Helps the File Translation Layer to optimize garbage collection
  - Introduced in between 2009/2011 in most OSs